Ashish **Sukumar**

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Professional Summary

Robotics Engineering M.S. student at WPI (GPA 3.83, May 2027) specializing in **computer vision, deep-learning perception, and multi-view 3D reconstruction**. Builds perception pipelines from scratch in **PyTorch** — NeRF-based 3D reconstruction, multi-scale CNN encoders, **camera calibration / pose estimation** via COLMAP SfM and bundle adjustment — plus **synthetic dataset generation in Blender** for training and evaluation. Strong foundation in probabilistic estimation (EKF, visual-inertial odometry) for **uncertainty-aware sensing**, with proven **real-world deployment** of autonomous systems on embedded hardware. Published researcher (ICIoT 2025) with prior R&D experience at e-Yantra, IIT Bombay. Seeking a Fall 2026 Directed Research / MS Thesis position in vision-based perception for safety-critical autonomous systems.

Education

Worcester Polytechnic Institute — **Worcester, MA**
M.S. Robotics Engineering — *2025–Present*
GPA: 3.83/4.00 Expected Graduation: May 2027
Relevant Coursework: Computer Vision (RBE 549), Deep Learning for Perception, Foundations of Robotics, Robot Dynamics, Motion Planning

SRM Institute of Science and Technology — **Chennai, India**
B.Tech Computer Science & Engineering — *2021–2025*
CGPA: 9.54/10 Excellence Award for Outstanding Student & Best Class Representative

Technical Skills

Languages: C++ (primary), Python, C, Bash
Deep Learning: PyTorch, TensorFlow, CNNs, Multi-scale Encoders, Bi-LSTM with Attention
Computer Vision & 3D: NeRF, SfM, COLMAP, Camera Calibration, Pose Estimation, Bundle Adjustment, Stereo Vision, Point Clouds
Estimation & Sensing: Visual-Inertial Odometry (S-MSCKF, EKF), Kalman Filtering, Covariance/Uncertainty Propagation, IMU Fusion, SLAM
Synthetic Data & Simulation: Blender (synthetic dataset generation), Gazebo, Webots, Genesis Simulator, RViz
Frameworks & Tools: ROS2, ROS, OMPL, MoveIt
Motion Planning: RRT, RRT*, PRM, PRM*, RRTConnect, AO-RRT, EST, SST, KPIECE1, TAMP / PDDL
Trajectory & Control: Kinodynamic Planning, Trajectory Optimization, Feedback Control, Dynamic Systems Modeling
Hardware: Arduino, Raspberry Pi, ESP32, IMU, Ultrasonic, IR, Proximity, Gas Sensors
Build & Systems: CMake, Docker, Git, Linux
Design Tools: Fusion 360, AutoCAD, TinkerCAD

Experience

Our Bright Future – STEMing Into Sports Program — **Nichols College, MA**
Robotics Instructor Intern — *Jun – Aug 7, 2026*

- Planned and facilitated hands-on **robotics, coding, and engineering lessons**; taught fundamental concepts and guided students through designing, building, and programming robotic projects.
- Mentored youth participants from diverse backgrounds, fostering **critical thinking, teamwork, and problem-solving**; monitored progress and supported program demonstrations and showcases.

e-Yantra, IIT Bombay — **Chennai, India**
Junior Project Technical Assistant — *Jun 2024–Feb 2025*

- Designed and implemented **autonomous task pipelines** and simulation environments in **Webots** for the national eYSRC competition, integrating **perception, decision-making, and embedded control** end-to-end.
- Built **image processing** and sensor-based modules (camera, IR, proximity) enabling real-world autonomous robot interaction and navigation.

- Performed **hardware-software debugging** across sensor, actuation, and control layers; produced technical documentation deployed nationally via mdBook.
- Formally recognised by **Prof. Kavi Arya** (PI, e-Yantra, IIT Bombay) for technical proficiency; conducted two workshops on Embedded Systems and Robotics for school students nationwide.

Robocare Lab, WPI **Worcester, MA**
Voluntary Research Assistant *Oct–Dec 2025*

- Contributed to **behavior design and perception pipelines** for the SoftBank **Pepper Robot**, integrating speech, gesture, and visual feedback via **ROS2**.
- Configured and tested onboard camera, microphone, and joint actuator systems; resolved behavioral failure modes through iterative lab testing.

WPI Small Business Digitization Initiative (SBDI) **Worcester, MA**
Digital Consultant *Oct–Dec 2025*

- Independently scoped, designed, and delivered end-to-end digital solutions for small business clients, managing full project lifecycle with minimal supervision.

Key Projects

NeRF: Built a complete **NeRF** pipeline from scratch in **PyTorch**: positional encoding, hierarchical coarse-fine sampling, and volume rendering. Achieved **27.42 dB PSNR / 0.9084 SSIM** (Lego) and 25.75 dB / 0.799 (Ship). Extended to custom real-world datasets using **COLMAP**-based SfM for camera pose extraction.

Deep VIO: Implemented full **S-MSCKF** pipeline (EKF, IMU propagation via RK4, stereo measurement updates, sliding-window camera states), achieving **RMSE ATE 0.12 m** on the EuRoC MAV benchmark after SE(3) alignment. Designed and trained three deep learning models (vision-only, IMU-only, visual-inertial gated fusion) using **multi-scale CNN encoders** and bi-directional LSTM with temporal attention. Built a **synthetic Blender VIO dataset** (20 scenes, 10K poses each, 100 Hz); applied pose graph optimization with loop-closure and smoothness constraints, reducing test ATE from 1.94 m to 1.18 m (**39% improvement**).

Fire Smart-Bot: Deployed an autonomous navigation system on custom embedded hardware with real-time **sensor fusion** (IR, gas, proximity), shortest-path planning, obstacle avoidance, BLE-based mobile control, and live video streaming. Validated in **real-world deployment**. Published at ICIoT 2025.

TAMP 7-DOF: Designed a full **TAMP** pipeline integrating **PDDL symbolic planning** with **OMPL motion planning**, enabling a 7-DOF robot arm to execute collision-aware motion primitives end-to-end. Implemented predicate extraction (ON, CLEAR, HOLDING, HANDEMPTY) and **closed-loop re-planning** for fault recovery.

RTP Planner: Implemented a complete **RTP** (Random Tree Planner) from scratch in **C++** using OMPL, supporting point and SE(2) square robots with custom collision checkers. Separately implemented **PRM, PRM*, and RRTConnect** in 7-DOF joint space with a self-collision checker; solved narrow-passage problems with targeted sampling. Benchmarked against EST and RRT on path optimality, computation time, and sampled states.

Kinodynamic: Implemented **AO-RRT** for dynamically constrained models including **car-like robots** (curvature limits, velocity bounds) and pendulum systems. Benchmarked **KPIECE1, EST, and SST** for convergence speed and path quality across varying system dynamics.

Robot Dynamics Simulator: Independent project. Built a full interactive **robot kinematics and dynamics simulator** (yamirobotdynamics.netlify.app) in vanilla JavaScript with **zero robotics-library dependencies**, implementing **Modern Robotics** (Lynch & Park) end-to-end: Product-of-Exponentials FK, **Levenberg-Marquardt IK** with joint-limit enforcement, space/body **Jacobians** via Adjoint maps, and **Recursive Newton-Euler** dynamics (gravity, Coriolis, mass matrix), analytically verified on a 3-DOF robot. Supports 2–6 DOF with live IK click-to-target and a real-time dynamics panel.

Publications

Ashish S., Jeya R., Rithish R., Donipart R., & Ganesh K. (2025). *Fire Detection and Risk Prediction for Smart Safety: A Neural Network-Driven IoT Approach.* IJERCSE.

Ashish S., Jeya R., Rithish R., Donipart R., & Ganesh K. (2025). *Fire Aware Smart-Bot with AI Responsive System.* ICIoT 2025.

Certifications & Achievements

- Red Hat Certified System Administrator (RHCSA)
- Oracle Cloud Infrastructure (OCI) Certified
- 2nd Place – Project Expo, SRMIST